Thomas Lee

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Technical Skills

Languages: C++, Python, JavaScript, GLSL
Tools: Git, Blender, Storyboard Pro 22, ComfyUI, PyTorch
Graphics: OpenGL, Vulkan, Unity, Maya, MotionBuilder, Blender

Education

University of Pennsylvania Masters of Computer Graphics and Game Technology - GPA: 4.0 May 2027
Boston University Bachelor of Arts and Science in Computer Science – GPA: 3.87 May 2025

Relevant Projects

Mini-Minecraft
Vulkan • C++ • Qt

- Built a Minecraft-style voxel engine in Vulkan, implementing a full texture atlas pipeline (staging buffers, image layouts, samplers, per-face UVs) and first-person player movement with voxel-accurate collision and ray-cast block interaction.
- Developed procedural water and weather rendering, including animated water surfaces and rain effects.

Monte Carlo Path Tracer
C++ • OpenGl • Qt

- Implemented physically-based path tracing with global illumination using Multiple Importance Sampling to combine light and BSDF sampling, reducing variance and improving convergence over naive methods.
- Supported diffuse, microfacet, and specular materials with correct throughput-based radiance accumulation.

CPU Rasterizer
C++ • Qt

- Built a CPU-only rasterizer with vertex transforms, triangle setup, depth buffering, clipping, and custom fragment shaders.

OpenGL Procedural Noise & Stylized Shaders
C++ • OpenGL • Qt

- Developed real-time procedural and stylized shaders, including noise-based materials and outline effects.

Half-Edge Mesh Editor & Catmull–Clark Subdivision Toolkit
C++ • Qt

- Developed a half-edge mesh framework with dynamic topology manipulation and Catmull–Clark subdivision.
- Designed an interactive editing interface for face/edge/vertex operations with live visual feedback.

3D Visualization of Insaniquarium
Python • PyOpenGL

- Simulated predator–prey interactions and flocking-inspired behavior.
- Applied physics-based potential fields to generate coherent, visually engaging motion.

Work Experience

AI Software Engineer Intern – Baobab Studios Burlingame, CA June 2025 – Sep 2025
Python • LoRA Training • ComfyUI Workflows • Kubernetes • Google Cloud

- Built an end-to-end generative pipeline for stylized character LoRA models, including dataset expansion, automated preprocessing, and model training.
- Integrated custom LoRAs into the studio's Kubernetes-based ComfyUI backend, enabling automated cloud inference and seamless return of generated frames to Storyboard Pro.
- Conducted ablation studies to evaluate LoRA variants and improve model quality for production workflows.

Software Engineer Intern – Baobab Studios Burlingame, CA June 2024 – Sep 2024
QtScript • Storyboard Pro 22 • Pipeline Tools

- Developed custom QtScript tools for Storyboard Pro 22 to automate and streamline storyboard artist workflows.
- Implemented features for pose capture, layer export, and automated image processing, significantly reducing repetitive manual steps.
- Collaborated with artists and production staff to identify workflow bottlenecks and deliver tools that improved speed, consistency, and ease of use.